Our model $A(\hat{n})$ = Fourier fit, term by term (worked)

Sources on the vision shell ($d = r$). $\psi \equiv \ell - \ell_0$, $c \equiv \cos b_0$, $\mu = c\cos\psi$, $\kappa = GM(<D)/(H_0^2 D^3)$.

Step 1. Expand the model in $\cos m\psi$

$$\frac{A}{A_{\rm all}} = N\Big[1 + \alpha\kappa(3\mu^2 - 1) + \alpha\kappa\tfrac{r}{D}\big(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3\big)\Big]$$

Identities: $\cos^2\psi = \frac{1}{2}(1+\cos 2\psi)$, $\cos^3\psi = \frac{1}{4}(3\cos\psi + \cos 3\psi)$

$$\mu^2 = \tfrac{c^2}{2}(1+\cos 2\psi), \qquad \mu^3 = \tfrac{c^3}{4}(3\cos\psi + \cos 3\psi)$$

$$3\mu^2 - 1 = \big(\tfrac{3}{2}c^2 - 1\big) + \tfrac{3}{2}c^2\cos 2\psi$$

$$\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3 = \tfrac{9}{2}c\cos\psi - \tfrac{15}{2}\cdot\tfrac{c^3}{4}(3\cos\psi + \cos 3\psi) = c\big(\tfrac{9}{2} - \tfrac{45}{8}c^2\big)\cos\psi - \tfrac{15}{8}c^3\cos 3\psi$$

$$\frac{A}{N} = \underbrace{1 + \alpha\kappa(\tfrac{3}{2}c^2 - 1)}_{m=0} + \underbrace{\alpha\kappa\tfrac{r}{D}c(\tfrac{9}{2} - \tfrac{45}{8}c^2)}_{m=1}\cos\psi + \underbrace{\alpha\kappa\tfrac{3}{2}c^2}_{m=2}\cos 2\psi \underbrace{-\alpha\kappa\tfrac{r}{D}\tfrac{15}{8}c^3}_{m=3}\cos 3\psi$$

90° sector averaging multiplies harmonic m by $W_m = \dfrac{\sin(m\pi/4)}{m\pi/4}$: $\quad W_1 = 0.9003,\ W_2 = 0.6366,\ W_3 = 0.3001.$

Step 2. Rotate the Fourier fit onto the model axis ($\ell_0 = 15°$, the m=2 peak)

$$A_{\rm fit} = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell$$

$c_m\cos m\ell + s_m\sin m\ell = C_m\cos m\psi + S_m\sin m\psi$, $\quad C_m = c_m\cos m\ell_0 + s_m\sin m\ell_0$, $\ S_m = -c_m\sin m\ell_0 + s_m\cos m\ell_0$

$$\begin{aligned}
C_1 &= 0.0296\cos 15° + 0.0434\sin 15° = 0.02859 + 0.01123 = +0.0398\\
C_2 &= 0.0636\cos 30° + 0.0366\sin 30° = 0.05508 + 0.01830 = +0.0734\\
C_3 &= -0.0299\cos 45° - 0.0114\sin 45° = -0.02114 - 0.00806 = -0.0292\\
S_1 &= -0.0296\sin 15° + 0.0434\cos 15° = -0.00766 + 0.04192 = +0.0343\\
S_2 &= -0.0636\sin 30° + 0.0366\cos 30° = -0.03180 + 0.03170 = -0.0001\\
S_3 &= +0.0299\sin 45° - 0.0114\cos 45° = +0.02114 - 0.00806 = +0.0131
\end{aligned}$$

Step 3. Set equal, harmonic by harmonic (model has only $\cos m\psi$ terms)

$$\begin{aligned}
m=0:&\quad 0.9965 = N\big[1 + \alpha\kappa(\tfrac{3}{2}c^2 - 1)\big]\\
m=1:&\quad C_1/W_1 = 0.0398/0.9003 = 0.04423 = N\alpha\kappa\tfrac{r}{D}\,c(\tfrac{9}{2} - \tfrac{45}{8}c^2)\\
m=2:&\quad C_2/W_2 = 0.0734/0.6366 = 0.11526 = N\alpha\kappa\,\tfrac{3}{2}c^2\\
m=3:&\quad C_3/W_3 = -0.0292/0.3001 = -0.09731 = -N\alpha\kappa\tfrac{r}{D}\tfrac{15}{8}c^3
\end{aligned}$$

Step 4. Solve (4 equations, 4 unknowns: $N,\ \alpha\kappa,\ r/D,\ c$**)**

(a) m=1 ÷ m=3 cancels $N\alpha\kappa\, r/D$ and one power of c:

$$\frac{0.04423}{-0.09731} = -0.4546 = \frac{\frac{9}{2} - \frac{45}{8}c^2}{-\frac{15}{8}c^2} \ \Rightarrow\ 0.8523\,c^2 = \tfrac{9}{2} - 5.625\,c^2 \ \Rightarrow\ c^2 = \frac{4.5}{6.4773} = 0.6947, \quad |b_0| = \arccos(0.8335) = 33.5°$$

(b) m=2 gives $N\alpha\kappa$:

$$N\alpha\kappa = \frac{0.11526}{\frac{3}{2}(0.6947)} = \frac{0.11526}{1.0421} = 0.1106$$

(c) m=3 gives r/D:

$$\frac{r}{D} = \frac{0.09731}{N\alpha\kappa\cdot\frac{15}{8}c^3} = \frac{0.09731}{0.1106\times 1.875\times 0.5790} = \frac{0.09731}{0.1201} = 0.810 \ \Rightarrow\ \frac{D}{r} = 1.234, \quad D = 1.234\times 14.3 = 17.6\ \text{Gpc}$$

(d) m=0 gives N, then $\alpha\kappa$:

$$N = 0.9965 - N\alpha\kappa(\tfrac{3}{2}c^2 - 1) = 0.9965 - 0.1106(0.0421) = 0.9918, \qquad \alpha\kappa = \frac{0.1106}{0.9918} = 0.1115$$

Check m=1: $\ 0.1106\times 0.810\times 0.8335\times(4.5 - 3.908)\times 0.9003 = 0.0398$ ✓

Step 5. What the model cannot match

An axisymmetric model has no $\sin m\psi$ terms, so $S_1 = +0.034$, $S_2 = -0.000$, $S_3 = +0.013$ are left over. S_1 is why the data dipole points at $\ell \approx 56°$ instead of along $\ell_0 = 15°$.

Caveats. ℓ_0 is set from the m=2 peak, not solved. Four equations in four unknowns match the C_m exactly by construction, so the test is whether the solved values are physical and how large the S_m are. The analytic form is a small-r/D expansion used at $r/D \approx 0.8$; the full numeric model gives $R^2 = 0.776$ at $D/r \approx 1.32$.